

Anqi Xu

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Senior Robotics & Perception Engineer with 15+ years building field-tested autonomy and production ML systems spanning robotics, AR/VR/MR, and applied AI. Experienced in building vision and ML systems that perceive, learn, and move atoms, not just bits, and accelerating deployment speed and robustness via agentic coding workflows.

WORK EXPERIENCE

Meta (Reality Labs)

02/2021 – Present

Robotics & Computer Vision Engineer

Redmond, WA

- Improved perceptual quality KPIs such as dynamic swim from -42% to +57% by product launch through the engineering of a **robotic and computer-vision test automation system** that tracked KPIs throughout 3 AR device programs' software builds via nightly CI-to-dashboard pipeline.

Tech: 3D localization, fiducial detection, world-locked rendering, sub-ms display-camera synchronization, 50-Gbps multi-camera ingestion, robot-arm control, OpenXR, C++, Python, Linux + Windows + Android, Hardware-In-The-Loop Continuous Integration (CI).

- Reduced calibration KPI such as augmentation error by 70% during AR prototype build via the deployment of an **automated factory calibration-validation pipeline** covering camera, display, and see-through calibration stages across 5 AR/MR programs to catch failures before devices ship.

Tech: Robotic camera capture, geometric camera & display calibration, perceptual metrics (augmentation error, binocular disparity error, lateral color separation), Gage Repeatability & Reproducibility (GR&R) analysis, OpenCV, Python, Linux + Windows + Android.

- Delivered software sprints in 2 weeks during typical 6-week factory builds by pioneering and training the team on **AI-native infrastructure**, authoring LLM skills to grade and debug calibration runs and to spawn goal-validated agentic workflows for feature development, all compatible with open-source and Meta harnesses.

Tech: Context and spec-driven engineering, agent team orchestration, across-harness skill/plugin development.

- Drove **team-wide Better Engineering initiatives** including modernizing data-capture infrastructure (legacy RPC → gRPC), personally removing 2M+ lines of dead code, enforcing hermetic builds, and improving test coverage from 23% to 68%, resulting in agent-friendly, fast iteration speeds and build times across the XR Calibration monorepo.

Tech: gRPC and protocol buffers, hermetic across-platform builds, agent-assisted development.

- Led a team of 4 engineers** and 3 Cross-Functional Partner teams to ship metrology systems on schedule across 5 years, by influencing org-level AR/MR product vision, translating into feature roadmaps, running architecture reviews and cross-discipline collaborations; mentored research interns resulting in 2 full-time IC conversions.

Element AI

03/2017 – 12/2020

Applied Research Scientist, Robotics Program Lead

Montreal, Canada

- Led a 4-person applied-research team, built a **reinforcement-learning-based autonomous sailing agent** that uncovered novel vessel control, contributing to Team ETNZ's 2020 America's Cup victory.

Tech: Deep Reinforcement Learning (PPO, SAC), Ray / Tune / RLlib, PyTorch, Docker, Python, AWS SageMaker, AWS S3.

- Co-published with a research intern an adversarial machine learning method for **robustness-testing SOTA visual trackers**, fooling them into losing track and locking onto inconspicuous billboards, applied to drone photography.

Tech: Adversarial ML, domain randomization, sim-to-real transfer, visual servo, Gazebo, ROS, quadrotor control, PyTorch, OpenCV.

- Prototyped a system for **localizing anomalies** on natural objects (e.g. apples) via instance segmentation, object-pose regression, and visual embeddings (PyTorch).

SKILLS

- Computer Vision: classical (feature detection, fiducial design, camera calibration, 3D geometry) and learning-based (object detection, segmentation, pose regression).

- Machine Learning: deep reinforcement learning, adversarial ML, probabilistic inference, attention / transformers, diffusion, foundation models (CNN, ViT, LLM/VLM).
- Robotics: localization, SLAM & VIO, planning, control; embedded software modules; electronics integration; hardware-in-the-loop; operational experience with UAVs, boats, and robot arms.
- AI-Native Engineering: context engineering, spec-driven development, skills & MCP, multi-agent orchestration.
- Research & Metrology: user-study design, robotics field experiments, statistical measurement analysis.

PUBLICATION HIGHLIGHTS

- R. Wiyatno and A. Xu. Physical Adversarial Textures that Fool Visual Object Tracking. ICCV 2019.
- A. Xu and G. Dudek. Maintaining Efficient Collaboration with Trust-Seeking Robots. IROS 2016. (Finalist, IROS KROS Best Paper Award on Cognitive Robotics)
- D. Meger *et al.* Learning Legged Swimming Gaits from Experience. ICRA 2015. (Finalist, Best Conference Paper Award)
- A. Xu and G. Dudek. OPTIMO: Online Probabilistic Trust Inference Model for Asymmetric Human-Robot Collaborations. HRI 2015.
- M. Doniec, A. Xu, and D. Rus. Robust Real-Time High Definition Underwater Video Streaming with AquaOptical II. ICRA 2013.

EDUCATION

Ph.D. in Computer Science — McGill University, Canada. 2017.

Thesis: Efficient Collaboration with Trust-Seeking Robots.

Vanier Canadian Graduate Scholar • CIPPRS Doctoral Dissertation Award 2016.

B.Eng. in Computer Engineering — McGill University, Canada. 2008.

Great Distinction • Dean's Honour List.

SIDE PROJECTS

- Conduit: **enabling fluid, on-the-go co-work** via an agentic assistant with modality-aware input corrections (e.g., Automatic Speech Recognition errors/typos) and adaptive output formatting (audio dialog, low-res AR displays), orchestrating cloud and local agent harnesses (Claude Code, OpenAI Codex, Google Antigravity, OpenCode).
- Bookkeeper: **automating financial reports and outlier detection** with a personal finance system that ingests merchant, processor, and payment sources, matches transactions to reconcile a SQLite ledger, categorizes and predicts labels for card transactions using finetuned BERT models, and serves analytics via FastAPI web client.
- Overcooked Bot: **building an interactive game-playing agent** via end-to-end perception, state-estimation, and control pipeline, using Faster-RCNN (Detectron2), Kalman tracking, homography-based grid detection, PyQt annotation GUI. [[target tracker](#), [grid detection](#) demos]
- Document Organizer: **automated filing of scanned documents** using local/cloud VLM to categorize, rename, and organize by visual and semantic content.
- [ueye_cam](#): **enabling synchronized high-performance camera capture** for robotics research with an open-source ROS driver for IDS uEye industrial cameras.

LEADERSHIP & COMMUNITY

- Program co-chair for [CRV 2018](#) and [CRV 2019](#).
- Reviewer for ICRA, IROS, RSS, RA-L, HRI, CoRL, CRV, ICCV, IJRR.